MOSFET

2	**MOSFET Amplifier**			
	2.1	MOSFET: Structure, working, characteristics; Two terminal basic MOS capacitor structure; non-ideal current voltage characteristics: finite output resistance, Body effect	**10**	**CO2**
	2.2	DC circuit analysis, load line and modes of operation and design of biasing circuits of MOSFET		
	2.3	Analysis of mid frequency CS amplifier using Hybrid π equivalent MOSFET circuits		
	2.4	**MOSFET as a Switch:** use of MOSFETs as a primary switching element in digital logic		

MOSFET

- FET
  - JFET: Junction Field Effect Transistor
  - MOSFET: Metal Oxide Semiconductor field Effect Transistor(used in design and construction of IC for digital computers)
    - Depletion
    - Enhancement
  - MESFET: Metal Semiconductor field Effect Transistor

Depletion Type MOSFET

Basic Construction of the n -channel

Depletion-type MOSFET

Depletion Type MOSFET

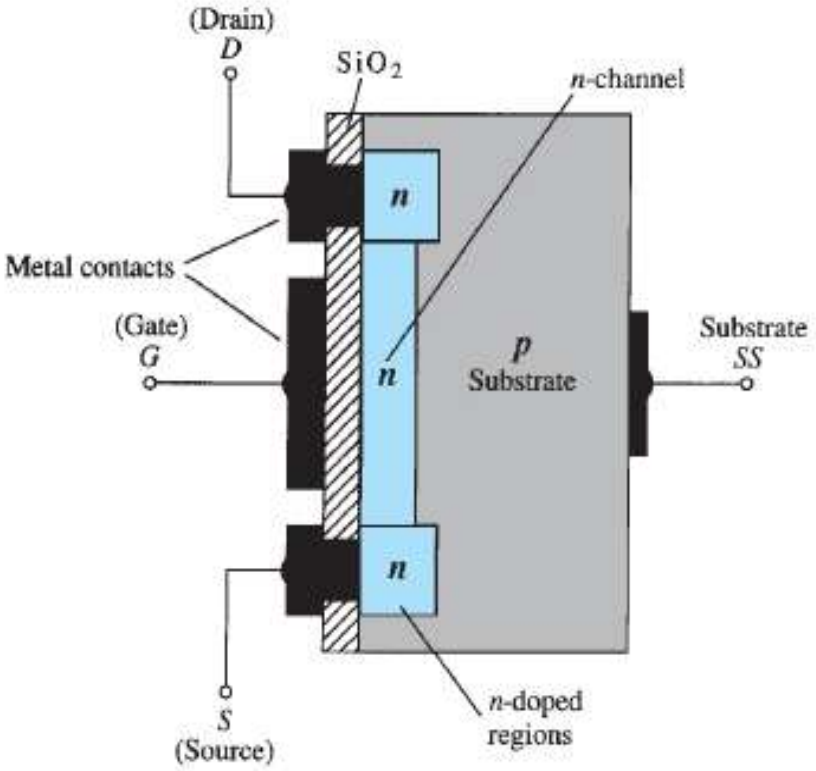
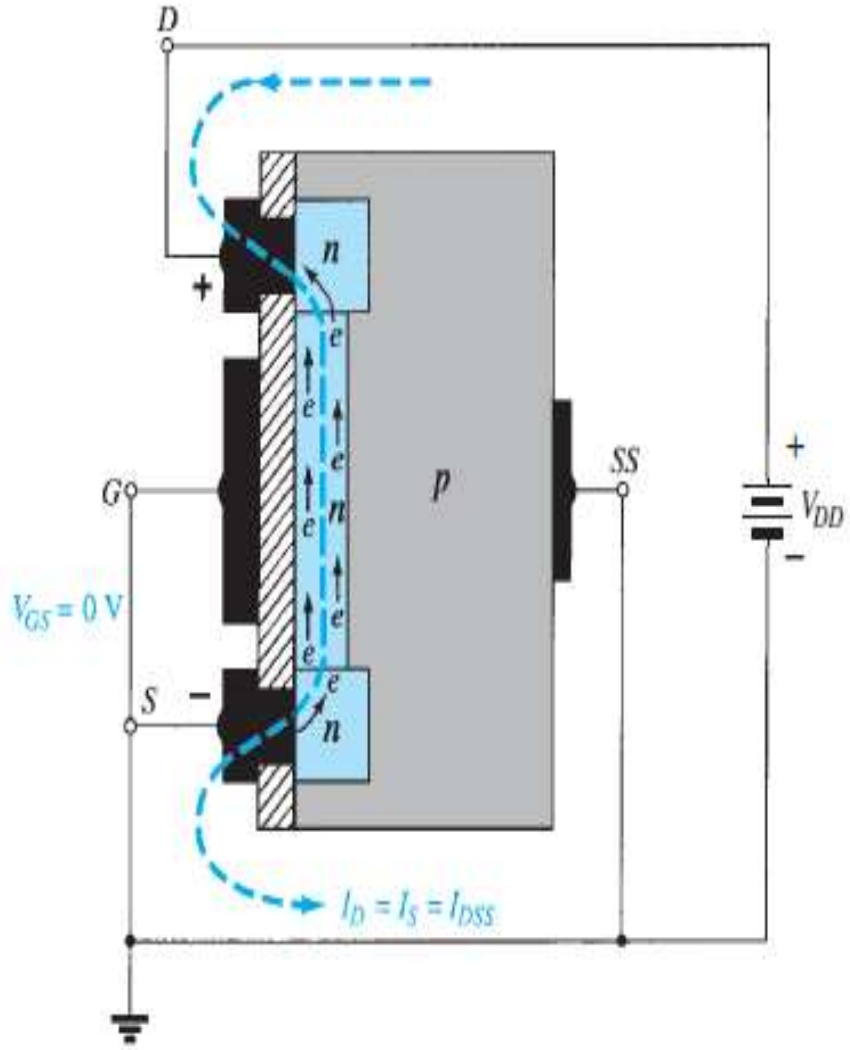


FIG. 6.24
n-Channel depletion-type MOSFET.

- A slab of *p* -type material is formed from a silicon base and is referred to as the *substrate. (*terminal labelled *SS)*
- The source and drain terminals are connected through metallic contacts to *n* -doped regions linked by an *n* –channel.
- The gate is also connected to a metal contact surface but remains insulated from the *n* -channel by a very thin silicon dioxide (SiO_2) layer.
- ***There is no direct electrical connection between the gate terminal and the channel of a MOSFET.***

Depletion Type MOSFET

- ***It is the insulating layer of SiO_2 in the MOSFET construction that accounts for the very desirable high input impedance of the device.***
- The reason for the label metal–oxide–semiconductor FET is now fairly obvious: *metal* for the drain, source, and gate connections; *oxide* for the silicon dioxide insulating layer; and *semiconductor* for the basic structure on which the *n* - and *p* -type regions are diffused.



- The gate-to-source voltage is set to 0 V by the direct connection from one terminal to the other, and a voltage V_{DD} is applied across the drain-to-source terminals.

- The result is an attraction of the free electrons of the *n*-channel for the positive voltage at the drain. The result is a current flowing in the channel.

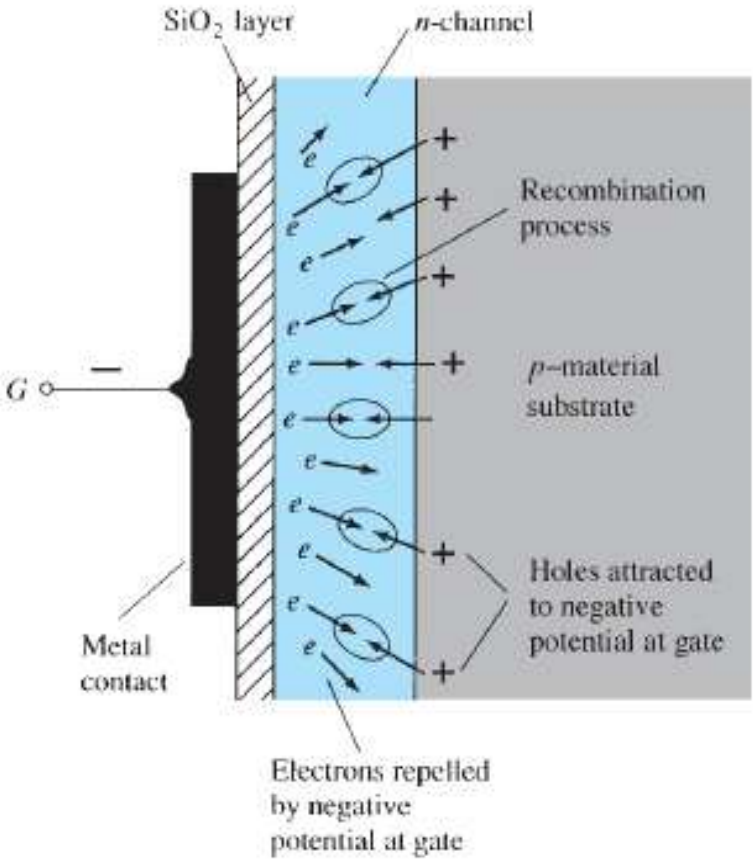


FIG. 6.27

Reduction in free carriers in a channel due to a negative potential at the gate terminal.

- V_{GS} is set at a negative voltage such as -1 V. The negative potential at the gate will tend to pressure electrons toward the *p* -type substrate (like charges repel) and attract holes from the *p* -type substrate (opposite charges attract).
- Depending on magnitude of the negative bias established by V_{GS} , a level of recombination between electrons and holes will occur that will reduce the number of free electrons in the *n* -channel available for conduction. The more negative the bias, the higher is the rate of recombination.
- The resulting level of drain current is therefore reduced with increasing negative bias for V_{GS} , as shown in Fig. 6.26 for V_{GS} = -1 V, -2 V, and so on, to the pinch-off level of -6 V.

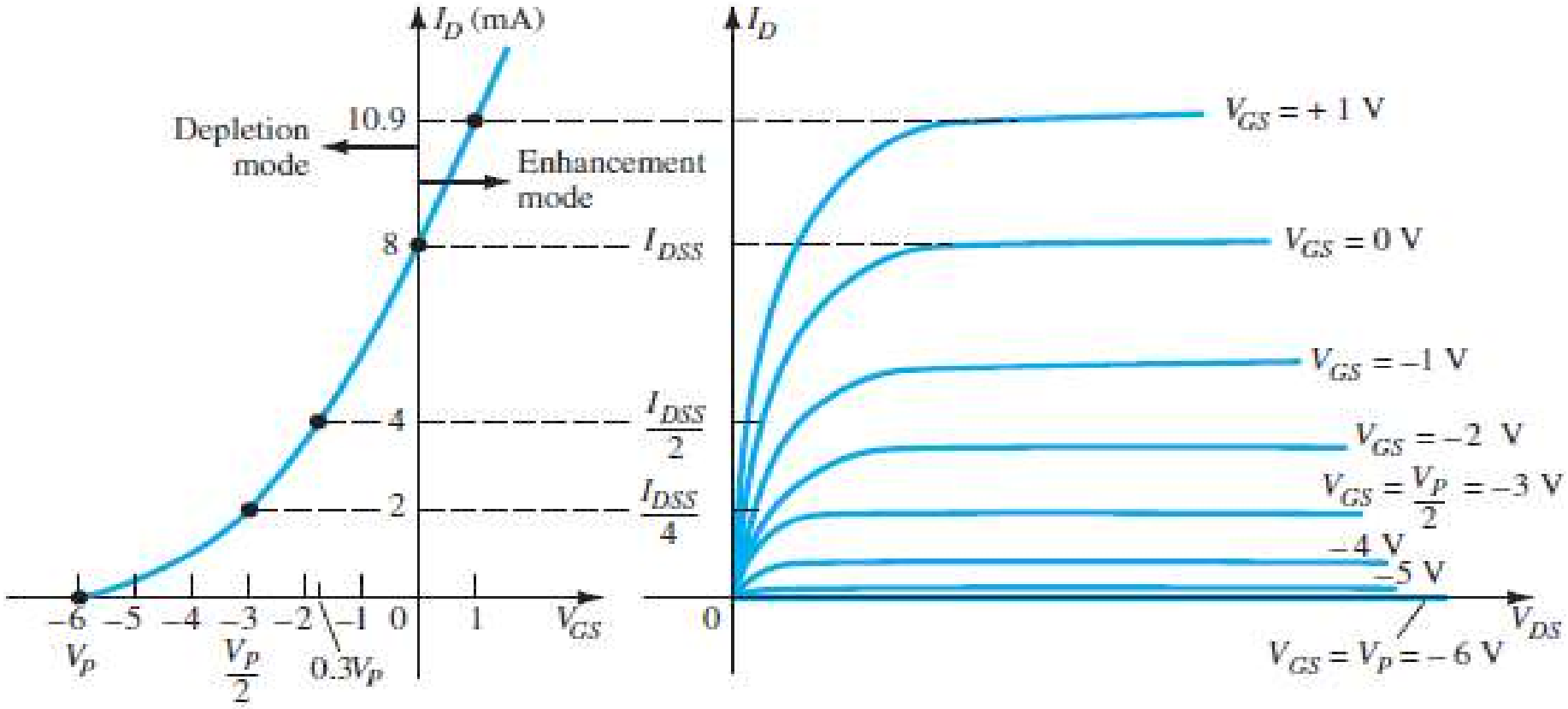


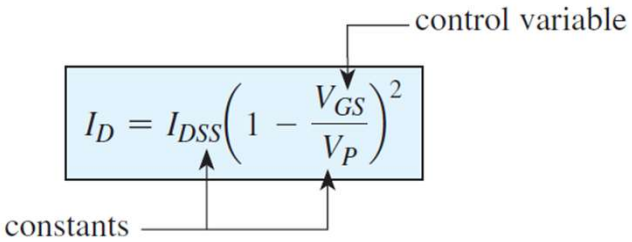
FIG. 6.26

Drain and transfer characteristics for an n-channel depletion-type MOSFET.

- For positive values of V_{GS} , the positive gate will draw additional electrons (free carriers) from the *p* -type substrate due to the reverse leakage current and establish new carriers through the collisions resulting between accelerating particles. As the gate-to-source voltage continues to increase in the positive direction.
- the application of a positive gate-to-source voltage has “enhanced” the level of free carriers in the channel compared to that encountered with V_{GS} =0 V. For this reason the region of positive gate voltages on the drain or transfer characteristics is often referred to as the *enhancement region* , with the region between cutoff and the saturation level of I_{DSS} referred to as the *depletion region.*
- It is particularly interesting and helpful that Shockley’s equation will continue

Shockley's equation

- The relationship between I_D and V_{GS} is defined by *Shockley's equation*



- Shockley's equation will continue to be applicable for the depletion-type MOSFET characteristics in both the depletion and enhancement regions. For both regions, it is simply necessary that the proper sign be included with V_{GS} in the equation and the sign be carefully monitored in the mathematical operations.

Sketch the transfer characteristics for an *n* -channel depletion-type MOSFET with I_{DSS} = 10 mA and V_P = -4 V.

$$\text{At } V_{GS} = 0\,\text{V}, \quad I_D = I_{DSS} = 10\,\text{mA}$$

$$V_{GS} = V_P = -4\,\text{V}, \quad I_D = 0\,\text{mA}$$

$$V_{GS} = \frac{V_P}{2} = \frac{-4\,\text{V}}{2} = -2\,\text{V}, \quad I_D = \frac{I_{DSS}}{4} = \frac{10\,\text{mA}}{4} = 2.5\,\text{mA}$$

$$I_D = \frac{I_{DSS}}{2},$$

$$V_{GS} = 0.3V_P = 0.3(-4\,\text{V}) = -1.2\,\text{V}$$

$$I_D = I_{DSS}\left(1 - \frac{V_{GS}}{V_P}\right)^2$$

$$= (10\,\text{mA})\left(1 - \frac{+1\,\text{V}}{-4\,\text{V}}\right)^2 = (10\,\text{mA})\,(1 + 0.25)^2 = (10\,\text{mA})\,(1.5625)$$

$$\cong 15.63\,\text{mA}$$

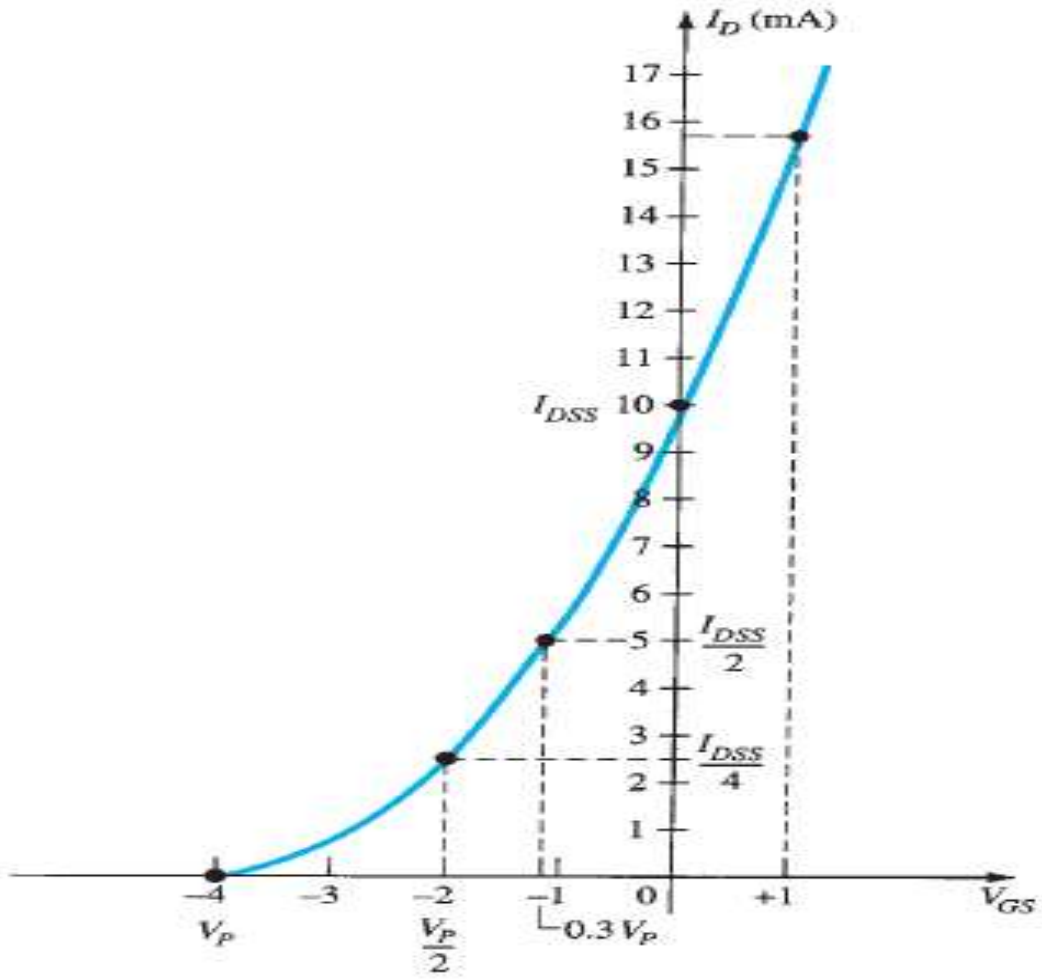


FIG. 6.28

Transfer characteristics for an n-channel depletion-type MOSFET with $I_{DSS} = 10$ mA and $V_P = -4$ V.

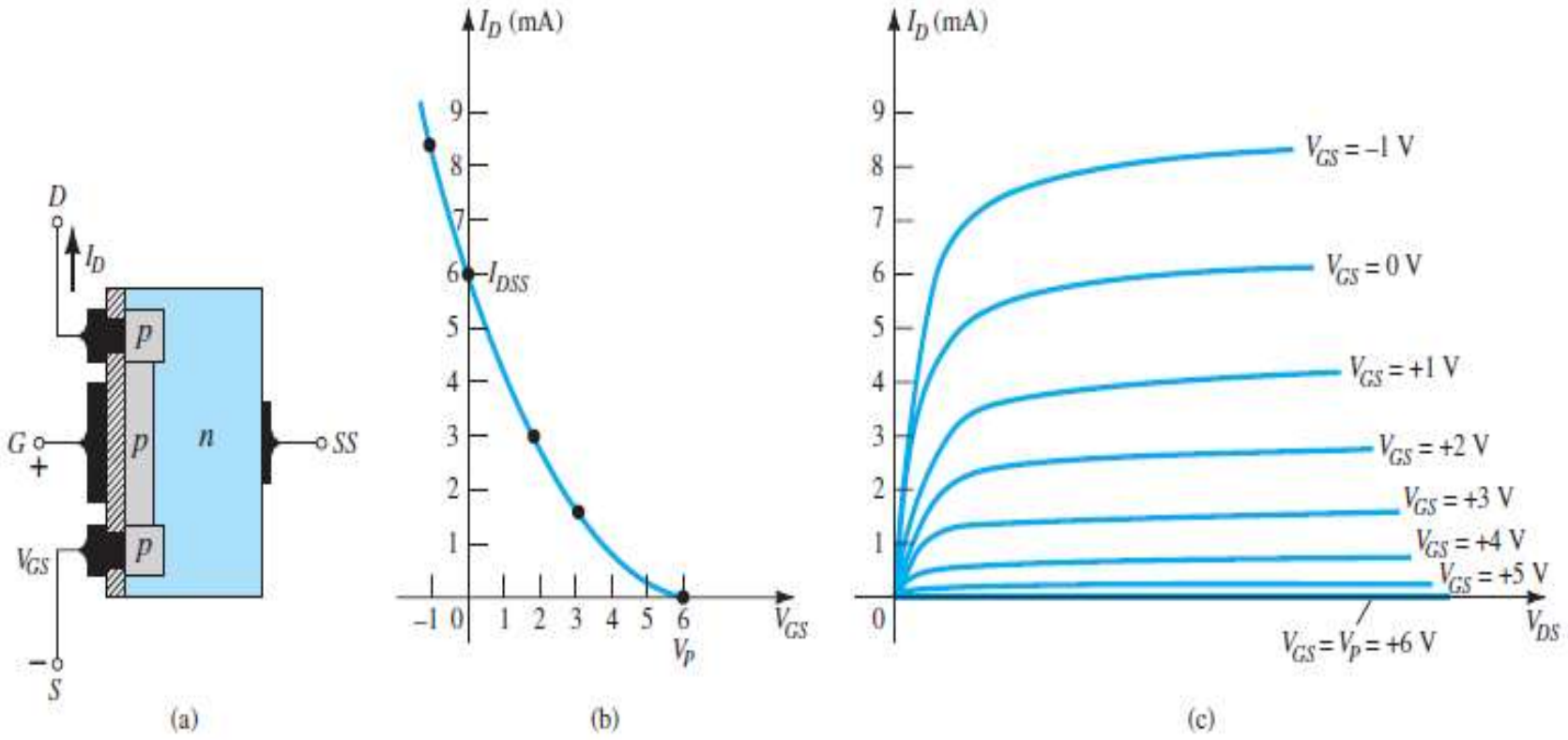


FIG. 6.29

p-Channel depletion-type MOSFET with $I_{DSS} = 6$ mA and $V_P = +6$ V.

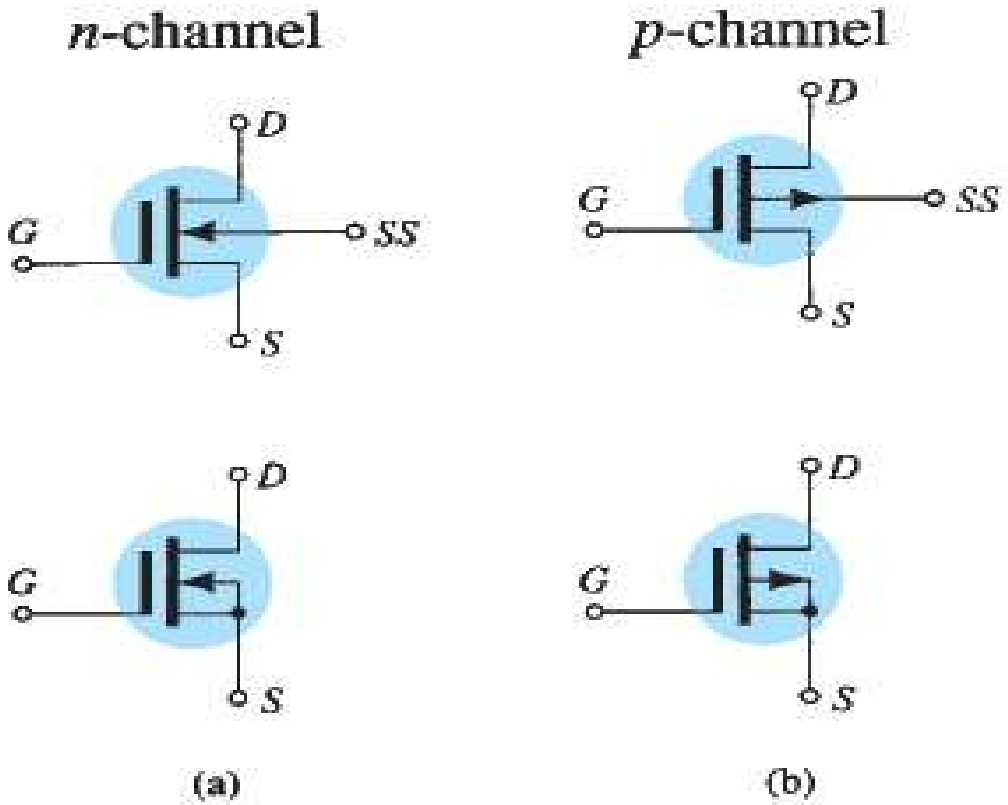


FIG. 6.30

Graphic symbols for: (a) n-channel depletion-type MOSFETs and (b) p-channel depletion-type MOSFETs.

Enhancement-type MOSFET

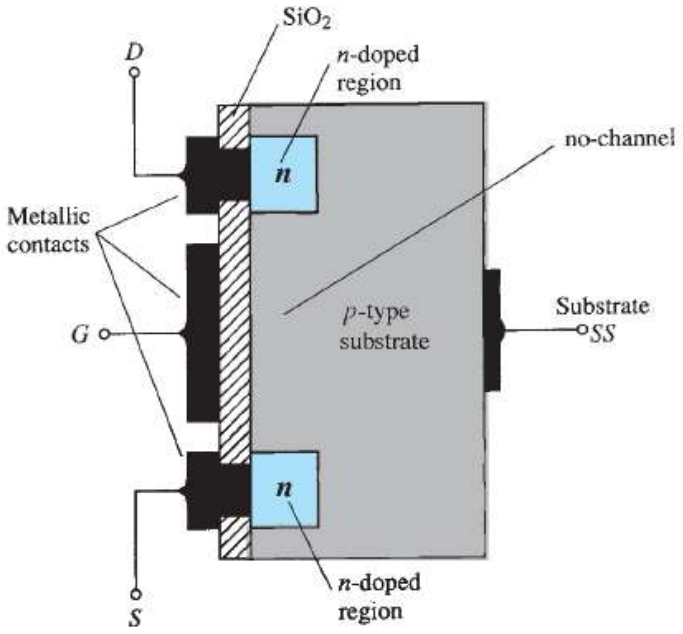


FIG. 6.32
n-Channel enhancement-type MOSFET.

The construction of an enhancement-type MOSFET is quite similar to that of the depletion-type MOSFET,

except

for the absence of a channel

between the drain and source terminals.

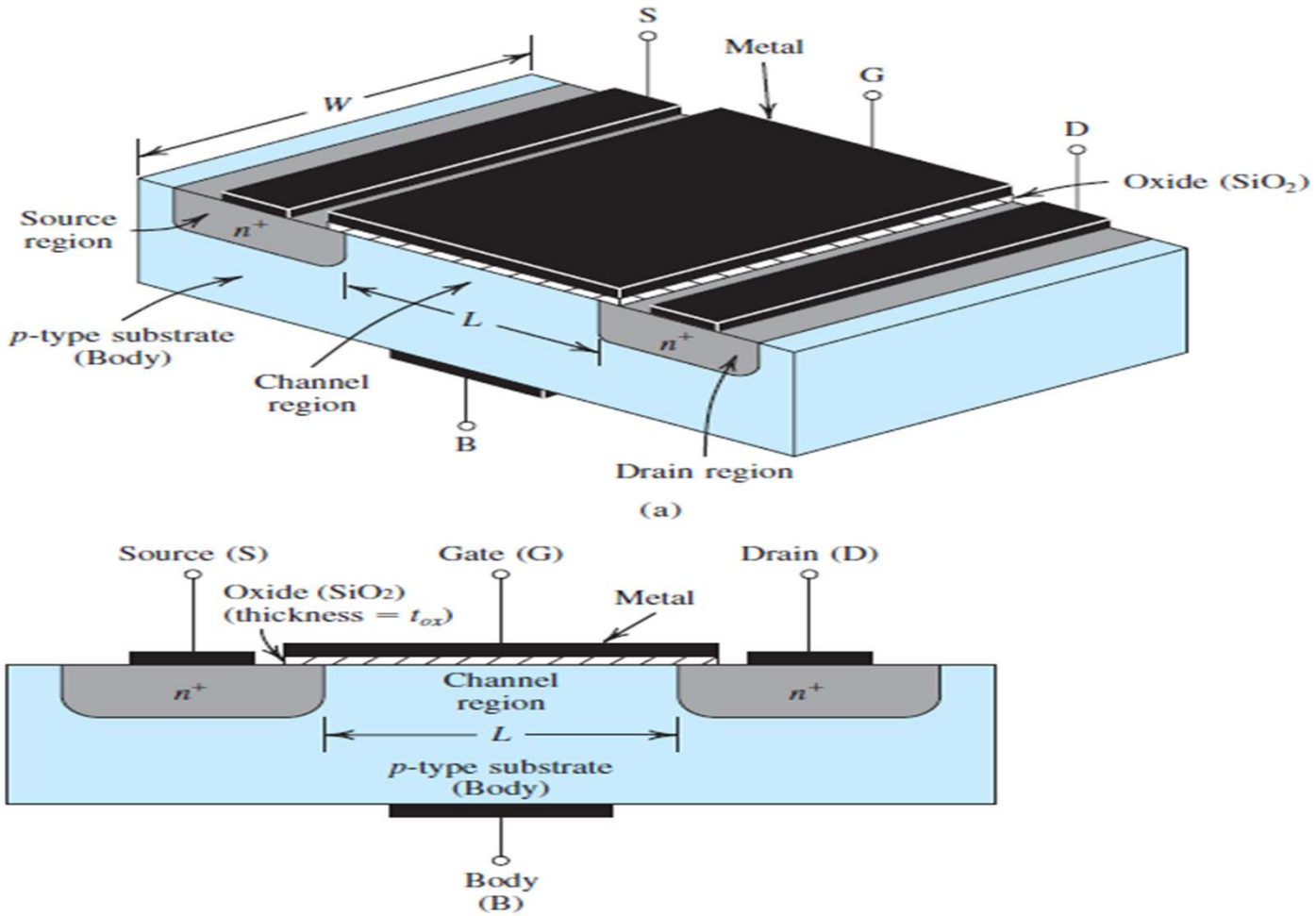


Figure 5.1 Physical structure of the enhancement-type NMOS transistor: **(a)** perspective view; **(b)** cross section. Typically $L = 0.03$ μm to 1 μm, $W = 0.1$ μm to 100 μm, and the thickness of the oxide layer (t_{ox}) is in the range of 1 to 10 nm.

Basic Operation and Characteristics

- If V_{GS} is set at 0 V and a voltage applied between the drain and the source of the device of Fig. 6.32 , the absence of an *n* -channel (with its generous number of free carriers) will result in a current of effectively 0 A—quite different from the depletion-type MOSFET and JFET, where $ID = I_{DSS}$.
- It is not sufficient to have a large accumulation of carriers (electrons) at the drain and the source (due to the *n* -doped regions) if a path fails to exist between the two.
- In Fig. 6.33 , both V_{DS} and V_{GS} have been set at some positive voltage greater than 0 V, establishing the drain and the gate at a positive potential with respect to the source.

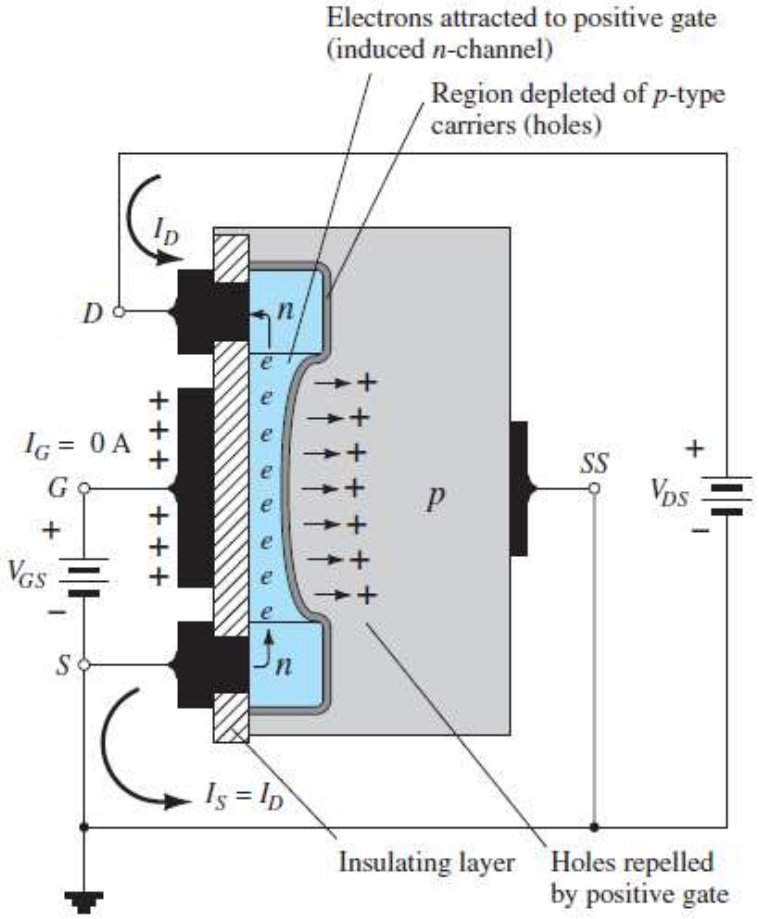


FIG. 6.33

Channel formation in the n-channel enhancement-type MOSFET.

The positive potential at the gate will pressure the holes (since like charges repel) in the *p* -substrate along the edge of the SiO_2 layer to leave the area and enter deeper regions of the *p* -substrate, as shown in the figure. The result is a depletion region near the SiO_2 insulating layer void of holes. However, the electrons in the *p* -substrate (the minority carriers of the material) will be attracted to the positive gate and accumulate in the region near the surface of the SiO_2 layer. The SiO_2 layer and its insulating qualities will prevent the negative carriers from being absorbed at the gate terminal

- The SiO_2 layer and its insulating qualities will prevent the negative carriers from being absorbed at the gate terminal. As V_{GS} increases in magnitude, the concentration of electrons near the SiO_2 surface increases until eventually the induced *n* -type region can support a measurable flow between drain and source. The level of V_{GS} that results in the significant increase in drain current is called the *threshold voltage* and is given the symbol V_T .
- On specification sheets it is referred to as V_{GS}(Th), although V_T is less unwieldy and will be used in the analysis to follow. Since the channel is nonexistent with V_{GS}= 0 V and "enhanced" by the application of a positive gate-to-source voltage, this type of MOSFET is called an *enhancement-type MOSFET.*

- As V_{GS} is increased beyond the threshold level, the density of free carriers in the induced channel will increase, resulting in an increased level of drain current. However, if we hold V_{GS} constant and increase the level of V_{DS}, the drain current will eventually reach a saturation level as occurred for the depletion-type MOSFET.
- The levelling off of I_D is due to a pinching-off process depicted by the narrower channel at the drain end of the induced channel as shown in Fig. 6.34 . Applying Kirchhoff's voltage law to the terminal voltages of the MOSFET of Fig. 6.34 , we find that

$$V_{DG} = V_{DS} - V_{GS}$$

$$V_{DG} = V_{DS} - V_{GS}$$

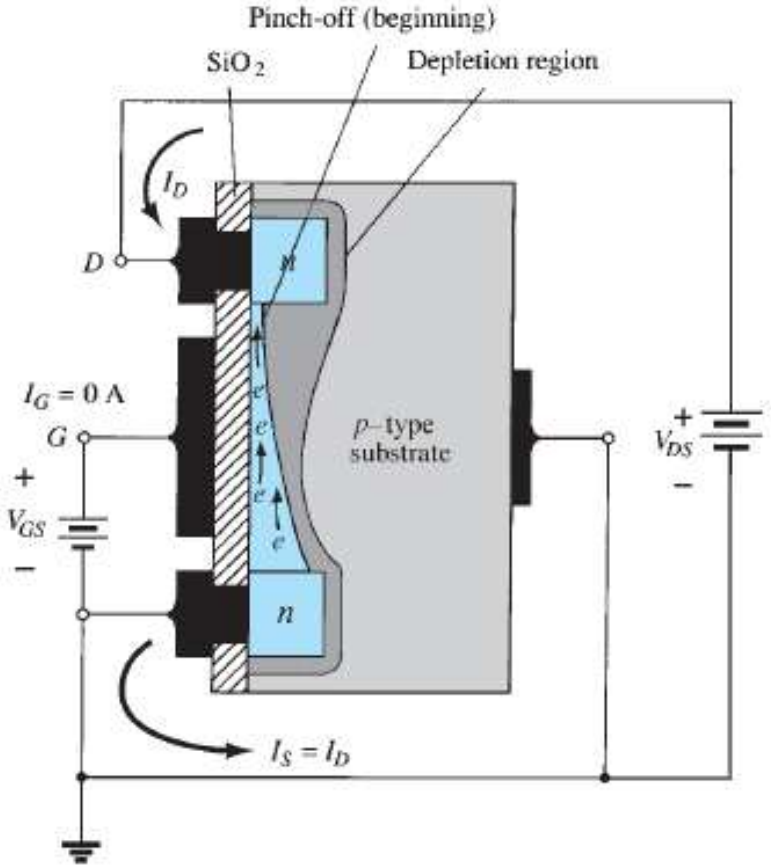


FIG. 6.34
Change in channel and depletion region with increasing level of V_{DS} for a fixed value of V_{GS}.

If V_{GS} is held fixed at some value such as 8 V and V_{DS} is increased from 2 V to 5 V, the voltage V_{DG} [by Eq. (6.13)] will increase from -6 V to -3 V and the gate will become less and less positive with respect to the drain. This reduction in gate-to-drain voltage will in turn reduce the attractive forces for free carriers (electrons) in this region of the induced channel, causing a reduction in the effective channel width. Eventually, the channel will be reduced to the point of pinch-off and a saturation condition will be established as described earlier for the depletion-type MOSFET.

In other words, any further increase in V_{DS} at the fixed value of V_{GS} will not affect the saturation level of I_D until breakdown conditions are encountered.

In fact, the saturation level for V_{DS} is related to the level of applied V_{GS} by

$$V_{DS_{sat}} = V_{GS} - V_T$$

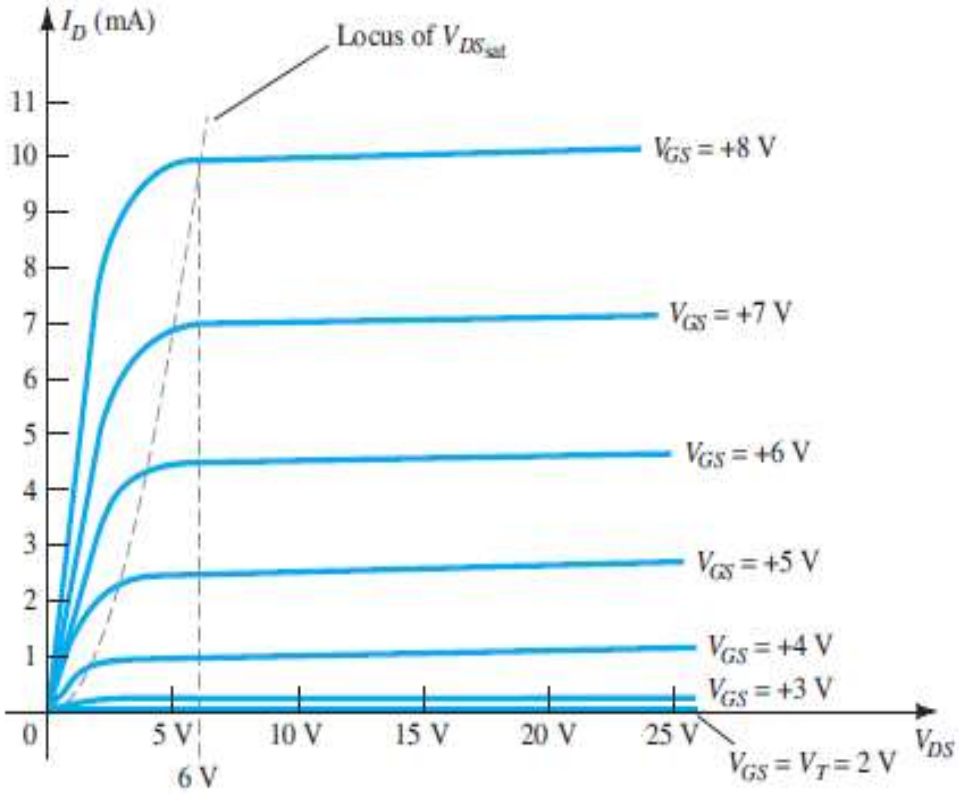


FIG. 6.35

Drain characteristics of an n-channel enhancement-type MOSFET with $V_T = 2$ V and $k = 0.278 \times 10^{-3}$ A/V^2.

Obviously, therefore, for a fixed value of V_T, the higher the level of V_{GS}, the greater is the saturation level for V_{DS}, as shown in Fig. 6.34 by the locus of saturation levels.

For the characteristics of Fig. 6.34, the level of V_T is 2 V, as revealed by the fact that the drain current has dropped to 0 mA. In general, therefore:

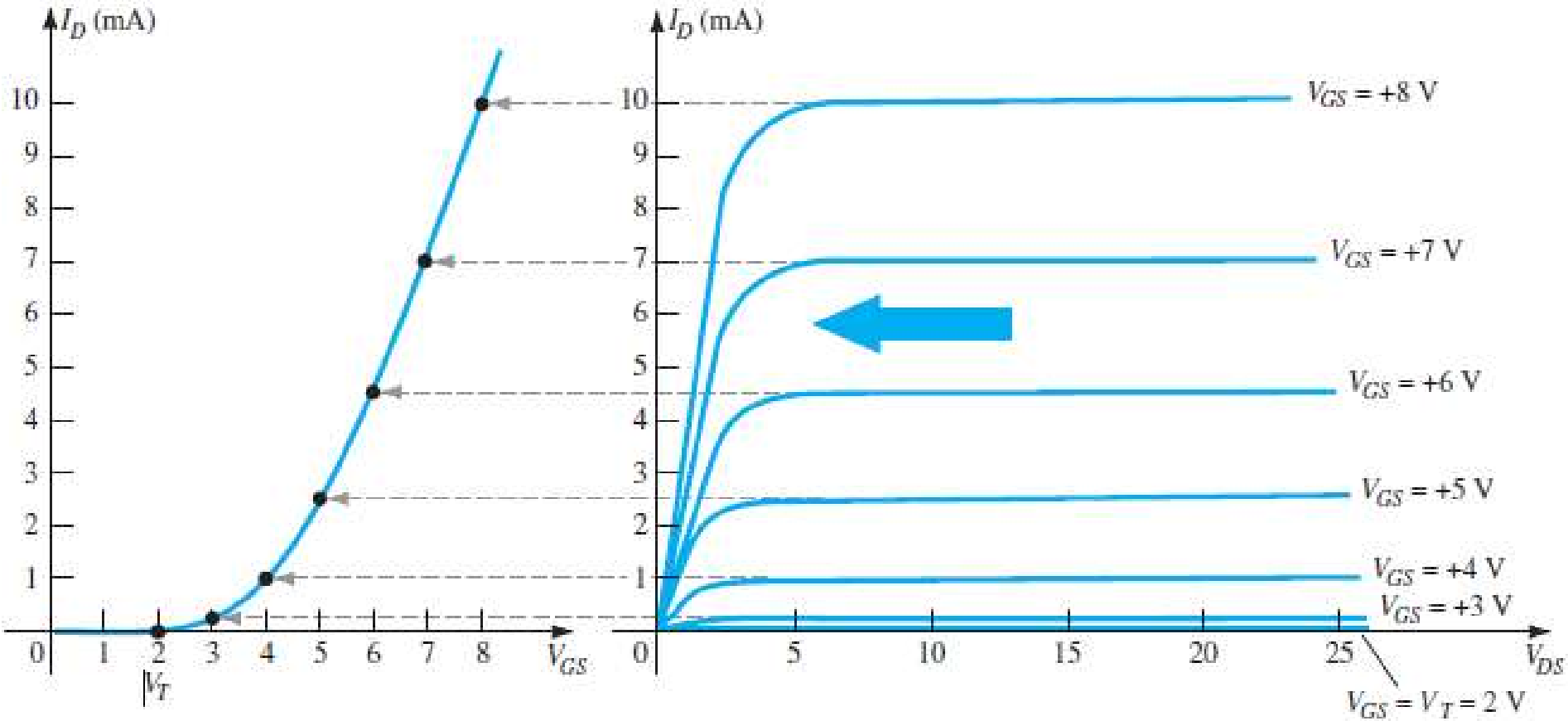
For values of V_{GS} less than the threshold level, the drain current of an enhancement type MOSFET is 0 mA.

- For levels of $V_{GS} > V_T$, the drain current is related to the applied gate-to-source voltage by the following nonlinear relationship:

$$I_D = k(V_{GS} - V_T)^2$$

Again, it is the squared term that results in the nonlinear (curved) relationship between I_D and V_{GS} . The k term is a constant that is a function of the construction of the device. The value of k can be determined from the following equation [derived from Eq. (6.15)], where $I_D(\text{on})$ and $V_{GS}(\text{on})$ are the values for each at a particular point on the characteristics of the device.

$$k = \frac{I_{D(\text{on})}}{(V_{GS(\text{on})} - V_T)^2}$$


I_D (mA)
10
9
8
7
6
5
4
3
2
1
0
1
2
3
4
5
6
7
8
V_{GS}
V_T
I_D (mA)
10
9
8
7
6
5
4
3
2
1
0
5
10
15
20
25
V_{DS}
V_{GS} = +8 V
V_{GS} = +7 V
V_{GS} = +6 V
V_{GS} = +5 V
V_{GS} = +4 V
V_{GS} = +3 V
$V_{GS} = V_T = 2$ V

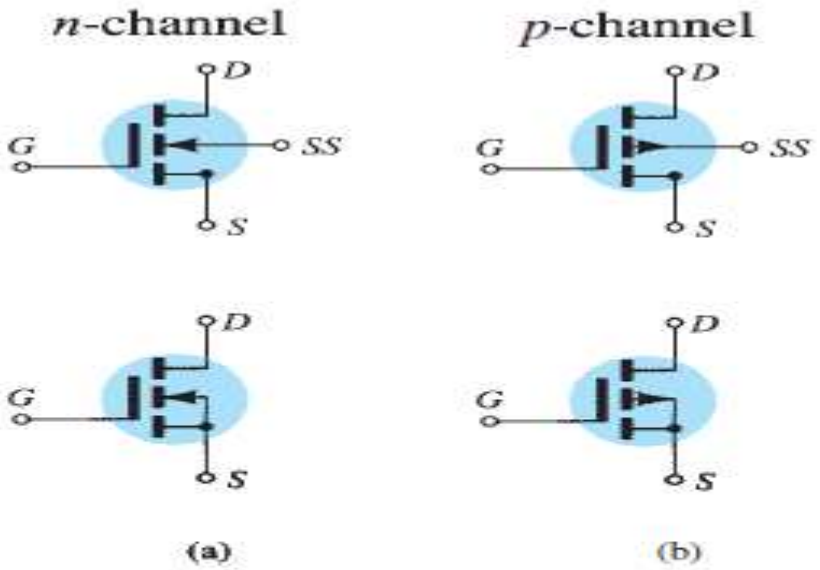
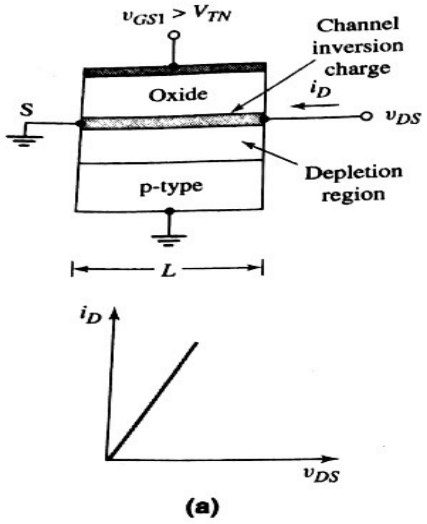
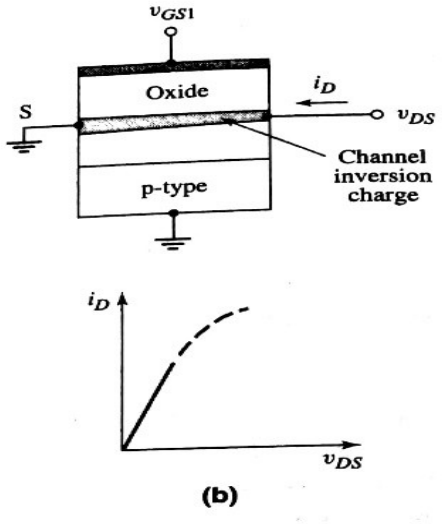
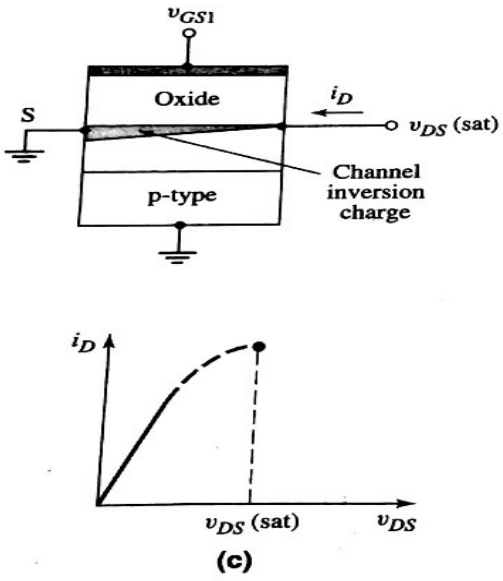


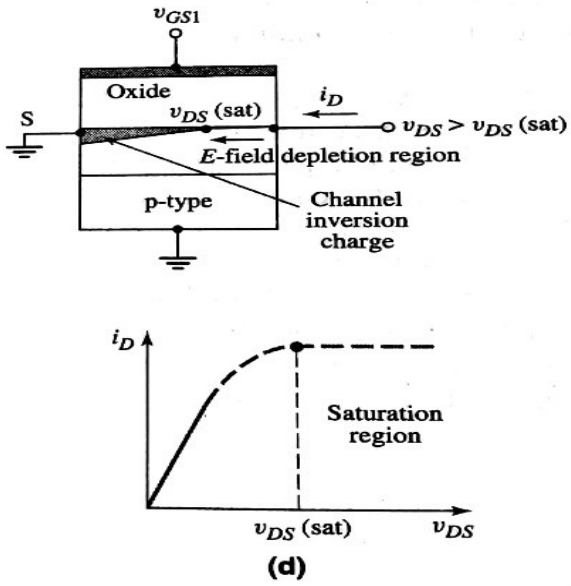
FIG. 6.39

Symbols for: (a) n-channel enhancement-type MOSFETs and (b) p-channel enhancement-type MOSFETs.









Cross Section and i_D versus V_{DS} curve for N channel enhancement MOSFET when $V_{GS} > V_{TN}$

a) A small V_{DS} value

b) A larger V_{DS} value for $V_{DS} < V_{DS}(sat)$

c) $V_{DS} = V_{DS}(sat)$

d) $V_{DS} > V_{DS}(sat)$

NMOS

The region for which $V_{DS} < V_{DS}(sat)$, is known as Non –Saturation region

$i_D = K_n [2(V_{GS} + V_{TN})V_{DS} - V^2_{DS}]$

In Saturation region, the ideal current–voltage characteristics

for $V_{DS} > V_{DS(SAT)}$, $i_D = K_n (V_{GS} - V_{TN})^2$

$K_n = W \mu n C_{OX} / 2 L$

Where C_{OX} is oxide capacitance per unit area $= \varepsilon_{OX}/t_{OX}$

t_{OX} is oxide thickness and

ε_{OX} is oxide permittivity $\varepsilon_{OX} = (3.9)(8.85 \times 10^{-14})$ F/cm

μ_n is mobility of electron in inversion layer

W is channel width

L is Channel length

PMOS

- $V_{SD}(sat) = V_{SG} + V_{TP}$
- Device biased in Non-saturation region

  $i_D = K_p [2(V_{SG} + V_{TP})V_{SD} - V^2_{SD}]$
- Device biased in saturation region

  $i_D = K_p (V_{SG} + V_{TP})^2$

NMOS

$V_{DS}(sat) = V_{GS} - V_{TN}$

Device biased in Non-saturation region

$$i_D = K_n[2(v_{GS} - V_{TN})v_{DS} - v_{DS}^2]$$

Device biased in saturation region for $V_{GS} > V_{TN}$,

$i_D = K_n (V_{GS} - V_{TN})^2$

Ex 3.1: Neamen

Ex 3.1: An NMOS transistor with $V_{TN} = 1$ V has a drain current $i_D = 0.8$ mA when $v_{GS} = 3$ V and $v_{DS} = 4.5$ V. Calculate the drain current when: (a) $v_{GS} = 2$ V, $v_{DS} = 4.5$ V; and (b) $v_{GS} = 3$ V, $v_{DS} = 1$ V.

$V_{DS\ (sat)} = V_{GS} - V_{TN}$

$= 3 - 1 = 2$ V $V_{DS} = 4.5$ V $> V_{DS\ (sat)}$

Transistor is in saturation region, So $i_D = K_n (V_{GS} - V_{TN})^2 = 0.8$ mA

Now Calculate Kn. Kn $= 0.2$ mA/V^2

a) MOSFET is in saturation $i_D = K_n (V_{GS} - V_{TN})^2 = 0.2(2-1)^2 = 0.2$ mA

b) MOSFET is in Non- saturation region

$$i_D = K_n[2(v_{GS} - V_{TN})v_{DS} - v_{DS}^2] = 0.6\text{mA}$$

EX 3.2: Neamen

Ex 3.2: (a) For a PMOS device, the threshold voltage is $V_{TP} = -2$ V and the applied source-to-gate voltage is $v_{SG} = 3$ V. Determine the region of operation when: (i) $v_{SD} = 0.5$ V; (ii) $v_{SD} = 2$ V; and (iii) $v_{SD} = 5$ V. (b) Repeat part (a) for a depletion-mode PMOS device with $V_{TP} = +0.5$ V. (Ans. (a) (i) nonsaturation, (ii) saturation, (iii) saturation; (b) nonsaturation, (ii) nonsaturation, (iii) saturation)

Test Your Understanding

TYU 3.1 (a) An n-channel enhancement-mode MOSFET has a threshold voltage of $V_{TN} = 1.2$ V and an applied gate-to-source voltage of $v_{GS} = 2$ V. Determine the region of operation when: (i) $v_{DS} = 0.4$ V; (ii) $v_{DS} = 1$ V; and (iii) $v_{DS} = 5$ V. (b) Repeat part (a) for an n-channel depletion-mode MOSFET with a threshold voltage of $V_{TN} = -1.2$ V. (Ans. (a) (i) nonsaturation, (ii) saturation, (iii) saturation; (b) (i) nonsaturation, (ii) nonsaturation, (iii) saturation)

TYU 3.2 The NMOS devices described in Exercise TYU3.1 have parameters $W = 100\,\mu$m, $L = 7\,\mu$m, $t_{ox} = 450$ Å, $\mu_n = 500$ cm^2/V–s, and $\lambda = 0$. (a) Calculate the conduction parameter K_n for each device. (b) Calculate the drain current for each bias condition. (Ans. (a) $K_n = 0.274$ mA/V^2 (b) $i_D = 0.132$, 0.175, and 0.175 mA; $i_D = 0.658$, 1.48, and 2.81 mA)

TYU 3.3 (a) A p-channel enhancement-mode MOSFET has a threshold voltage of $V_{TP} = -1.2$ V and an applied source-to-gate voltage of $v_{SG} = 2$ V. Determine the region of operation when (i) $v_{SD} = 0.4$ V, (ii) $v_{SD} = 1$ V, and (iii) $v_{SD} = 5$ V. (b) Repeat part (a) for a p-channel depletion-mode MOSFET with a threshold voltage of $V_{TP} = +1.2$ V. (Ans. (a) (i) nonsaturation, (ii) saturation, (iii) saturation; (b) (i) nonsaturation, (ii) nonsaturation, (iii) saturation)

Ref: Neaman

TYU 3.4 The PMOS devices described in Exercise TYU3.3 have parameters $W = 40\,\mu$m, $L = 2\,\mu$m, $t_{ox} = 350$ Å, and $\mu_p = 300\ \text{cm}^2/\text{V–s}$. (a) Calculate the conduction parameter K_p for each device. (b) Calculate the drain current for each bias condition described in Exercise TYU3.3. (Ans. (a) $K_p = 0.296\ \text{mA/V}^2$, (b) (i) 0.142 mA, (ii) 0.189 mA, (iii) 0.189 mA; (b) (i) 0.710 mA, (ii) 1.60 mA, (iii) 3.03 mA)

TYU 3.5 For an NMOS enhancement-mode device, the parameters are: $V_{TN} = 0.8\,\text{V}$ and $K_n = 0.1\ \text{mA/V}^2$. The device is biased at $v_{GS} = 2.5\,\text{V}$. Calculate the drain current when $v_{DS} = 2\,\text{V}$ and $v_{DS} = 10\,\text{V}$ for: (a) $\lambda = 0$ and (b) $\lambda = 0.02\,\text{V}^{-1}$. (c) Calculate the output resistance r_o for parts (a) and (b). (Ans. (a) $i_D = 0.289\,\text{mA}$ for both 2 and 10 V; (b) $i_D = 0.30\,\text{mA}$ (2V), $i_D = 0.347\,\text{mA}$ (10 V); (c) $r_o = \infty$ (a), $r_o = 173\,\text{k}\Omega$ (b))

TYU 3.6 An NMOS transistor has parameters $V_{TNO} = 1\,\text{V}$, $\gamma = 0.35\,\text{V}^{1/2}$, and $\phi_f = 0.35\,\text{V}$. Calculate the threshold voltage when: (a) $v_{SB} = 0$, (b) $v_{SB} = 1\,\text{V}$, and (c) $v_{SB} = 4\,\text{V}$. (Ans. (a) 1 V, (b) 1.16 V, (c) 1.47 V)

EXAMPLE 3.3

Objective: Calculate the drain current and drain-to-source voltage of a common-source circuit with an n-channel enhancement-mode MOSFET. Find the power dissipated in the transistor.

For the circuit shown in Figure 3.25(a), assume that $R_1 = 30\,\text{k}\Omega$, $R_2 = 20\,\text{k}\Omega$, $R_D = 20\,\text{k}\Omega$, $V_{DD} = 5\,\text{V}$, $V_{TN} = 1\,\text{V}$, and $K_n = 0.1\,\text{mA/V}^2$.

$$V_G = V_{GS} = \left(\frac{R_2}{R_1 + R_2}\right) V_{DD}$$

Calculate I_D

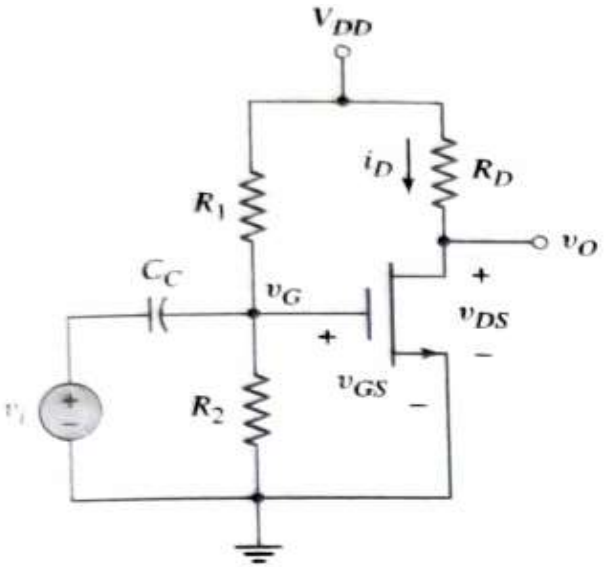


Figure 3.24 An NMOS common-source circuit

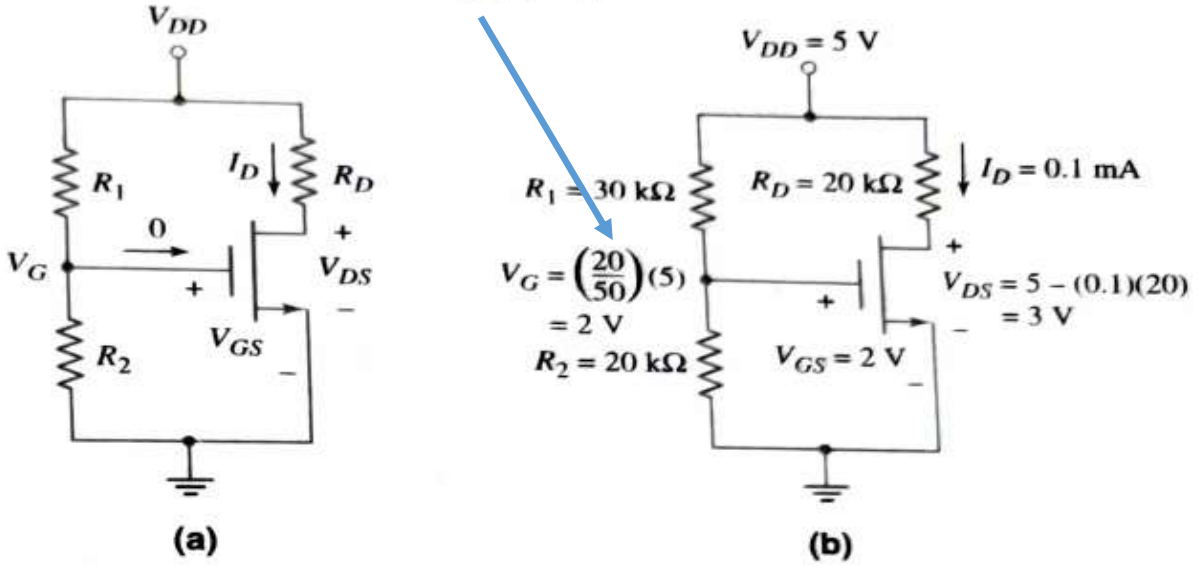


Figure 3.25 (a) The dc equivalent circuit of the NMOS common-source circuit and (b) the NMOS circuit for Example 3.3, showing current and voltage values

Assuming that the gate-to-source voltage given by Equation (3.12) is greater than V_{TN}, and that the transistor is biased in the saturation region, the drain current is

$$I_D = K_n(V_{GS} - V_{TN})^2 \tag{3.13}$$

$$I_D = 0.1(2-1)^2$$

The drain-to-source voltage is

$$V_{DS} = V_{DD} - I_D R_D \tag{3.14}$$

$$V_{DS} = V_{DD} - I_D R_D = 5 - (0.1)(20) = 3\ V$$

$V_{DS\,(SAT)} = 2-1 = 1V$

$3\ V > 1V$

If $V_{DS} > V_{DS}(\text{sat}) = V_{GS} - V_{TN}$, then the transistor is biased in the saturation region, as we initially assumed, and our analysis is correct. If $V_{DS} < V_{DS}(\text{sat})$, then the transistor is biased in the nonsaturation region, and the drain current is given by the more complicated characteristic Equation (3.2(a)).

The power dissipated in the transistor, since there is no gate current, is simply given by

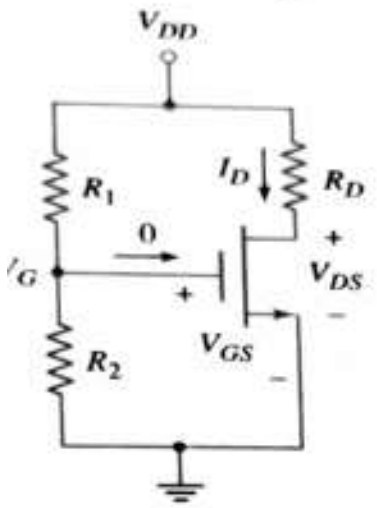
$$P_T = I_D V_{DS} \tag{3.15}$$

Ex 3.3: The transistor in Figure 3.25(a) has parameters $V_{TN} = +2\,\text{V}$ and $K_n = 0.25\,\text{mA/V}^2$. The circuit parameters are $V_{DD} = 10\,\text{V}$, $R_1 = 280\,\text{k}\Omega$, $R_2 = 160\,\text{k}\Omega$, and $R_D = 10\,\text{k}\Omega$. Find I_D, V_{DS}, and the power dissipated in the transistor. (Ans. $I_D = 0.669\,\text{mA}$, $V_{DS} = 3.31\,\text{V}$, $P = 2.21\,\text{mW}$)

$$V_G = V_{GS} = \left(\frac{R_2}{R_1 + R_2}\right) V_{DD}$$

$V_G = 3.64$ V

$V_{GS} = V_G = 3.64$



$$I_D = K_n(V_{GS} - V_{TN})^2$$

$I_D = 0.669$ mA

$V_{DS} = V_{DD} - I_D R_D$

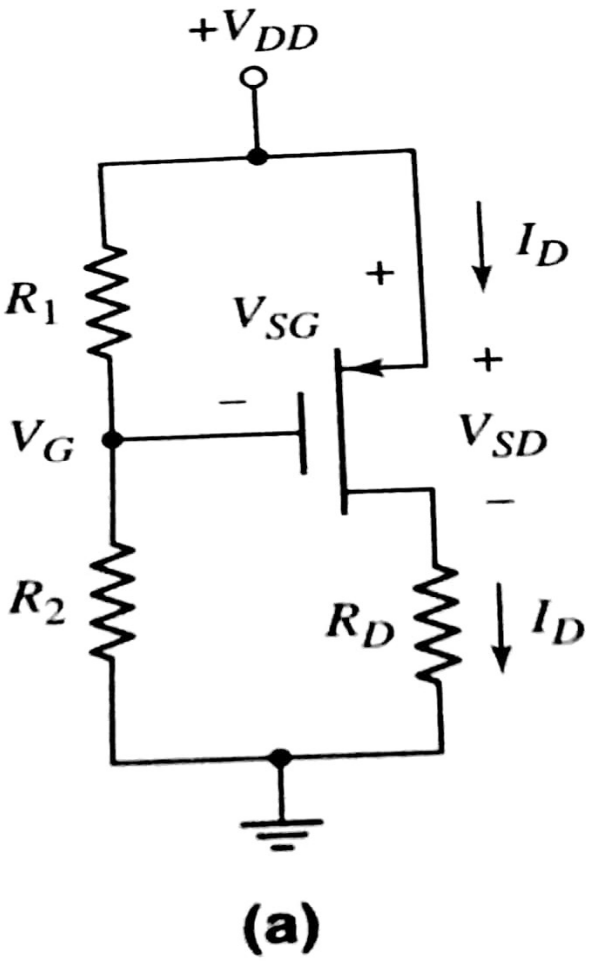
$= 10 - 6.7 = 3.31$ V

If $V_{DS} > V_{DS}(\text{sat}) = V_{GS} - V_{TN}$, then the transistor is biased in the saturation region,

$P = I_D V_{DS}$

$= 2.21$ mW

PMOS Common source circuit



(a)

V_{DD} = 5 V; R_1 = R_2 = 50 K Ω; R_D = 7.5 K Ω; V_{TP} = -0.8 V

I_D = ? V_{SD}= ?

Kp = 0.2 mA/V^2

$$V_G = \frac{R_2}{R_1 + R_2} V_{DD}$$

V_{SG}= V_{DD} – V_G

Assuming transistor is in sa

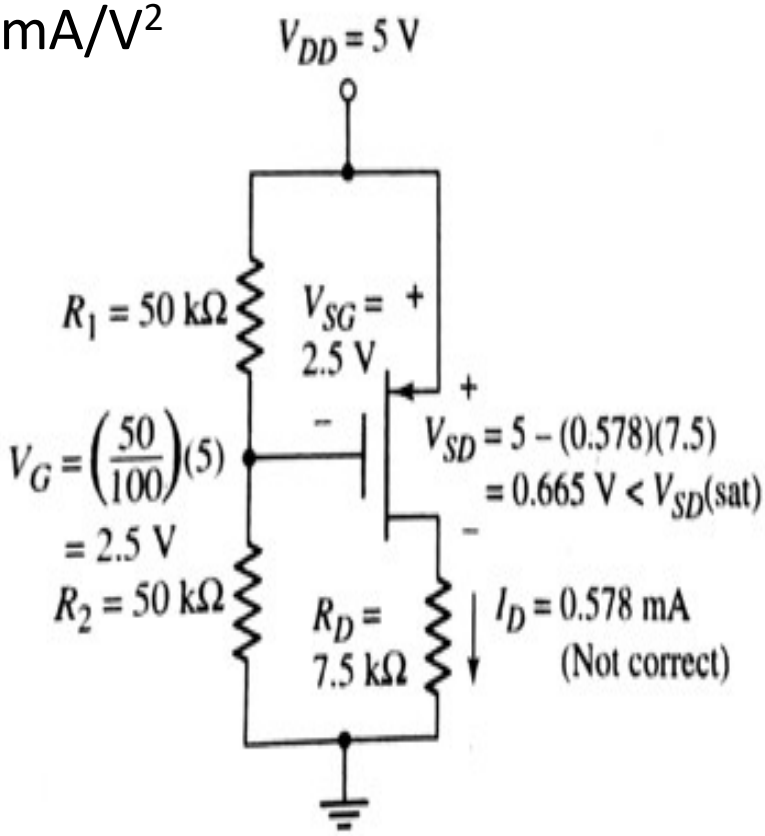
$i_D = K_p (V_{SG} + V_{TP})^2$

V_{SD}= V_{DD} - I_D R_D

$V_{SD\,(SAT)}$ = V_{SG} + V_{TP}

Is V_{SD} > V_{SD}(SAT) ? No.

Assumption is wrong.



$V_{DD} = 5\ V;\ R_1 = R_2 = 50\ K\ \Omega;\ R_D = 7.5\ K\ \Omega;\ V_{TP} = -0.8\ V$

- Transistor must be in non-saturation region

$I_D = K_p [2(V_{SG} + V_{TP})V_{SD} - V^2_{SD}]$

$V_{SD} = V_{DD} - I_D R_D$

$I_D = K_p [2(V_{SG} + V_{TP})(V_{DD} - I_D R_D) - (V_{DD} - I_D R_D)^2]$

$I_D = -11.25\, I_D^{\,2} + 9.9\, I_D - 1.6$

$0 = -11.25\, I_D^{\,2} + 8.9\, I_D - 1.6$

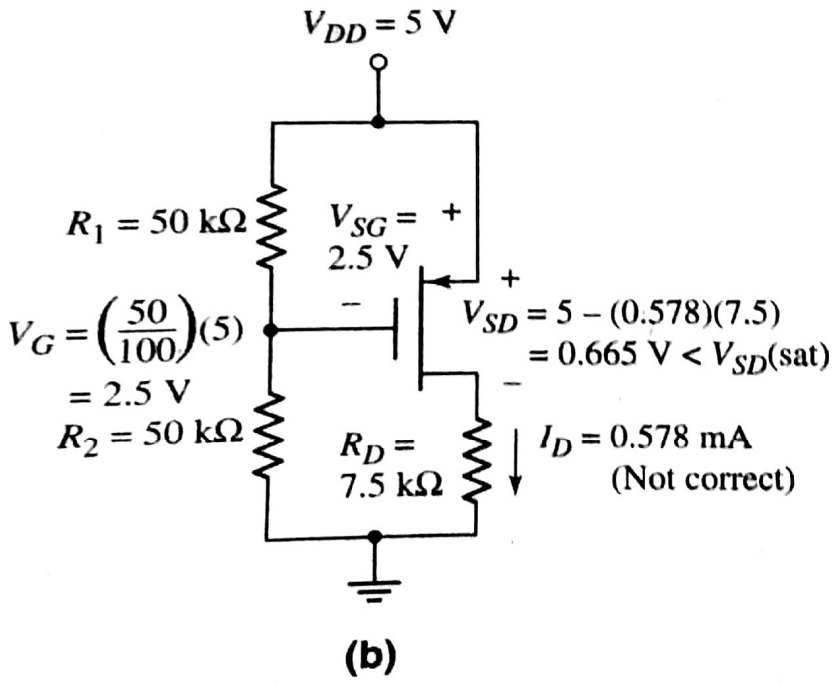
If $\boldsymbol{I_{D\,1} = 0.515}$, then $\boldsymbol{V_{SD}} = V_{DD} - I_D R_D = \mathbf{1.14\ V}$; $V_{SD(SAT)} = V_{SG} + V_{TP} = 1.7\ V$

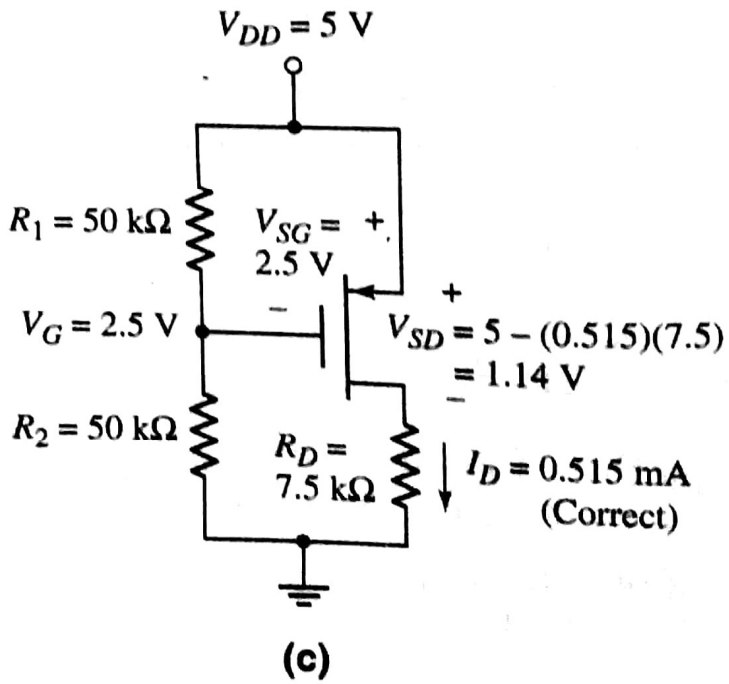
Is $V_{SD} < V_{SD}(SAT)$? Yes. So, Assumption is correct. Transistor must be in non-saturation region

If $I_{D\,2} = 0.276$, then $V_{SD} = V_{DD} - I_D R_D = 2.93\ V$; Is $V_{SD} < V_{SD}(SAT)$? Not correct. (As assumption is transistor is in nonsaturation)

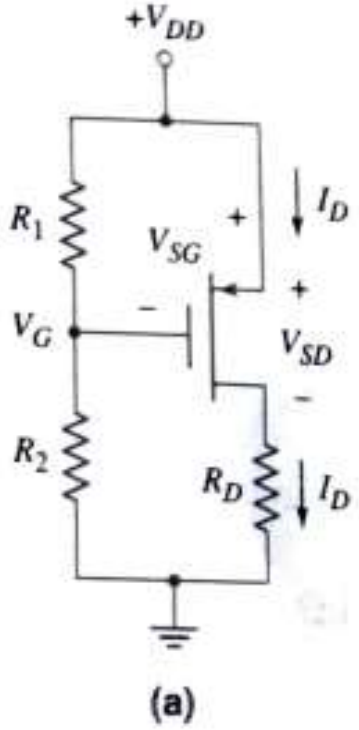
So, $\boldsymbol{I_{D\,1} = 0.515\ mA,\ V_{SD} = 1.14\ V}$

DC Equivalent Circuit





Ex 3.4: The transistor in Figure 3.26(a) has parameters $V_{TP} = -1.2\,\text{V}$ and $K_p = 0.4\,\text{mA/V}^2$. The circuit is biased at $V_{DD} = 10\,\text{V}$. Assume $R_1 \| R_2 = 200\,\text{k}\Omega$. Design the circuit such that $I_{DQ} = 1.2\,\text{mA}$ and $V_{SDQ} = 4\,\text{V}$. (Ans. $R_1 = 283\,\text{k}\Omega$, $R_2 = 682\,\text{k}\Omega$, $R_D = 5\,\text{k}\Omega$)



(a)

$I_D = Kp(Vsg + V_{TP})^2$

$1.2 = 0.4 (V_{SG} + (-1.2))^2$

$V_{SG} = (R_1/ (R_1+R_2)) V_{DD} = 2.93$ V

$2.93 = (1/ R_2) 200 (10)$

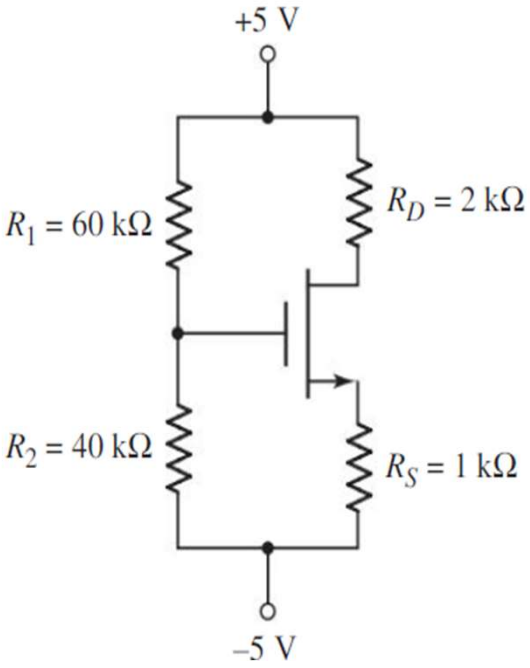
$R_2 = 682$ K

$200 = 682R_1/ (682+ R_1)$

$R_1 = 283$ K

$R_D = (10-4) / I_D = 6/1.2 = 5$K

Ex 3.5: (a) For the transistor in the circuit in, the parameters are VTN = 1V and Kn = 0.5 mA/V2. Determine VGs, ID, and VDs. (b) Determine the variation in the Q-point values for a ±5 percent variation in Kn and VTN.
(Ans. (a) VGS=2.65V, ID =1.35 mA, VDS = 5.94 V; (b)1.297 ≤ I_{DQ} ≤1.411 mA, 5.768≤V_{DSQ} ≤6.108V)



Assume transistor biased in saturation region Calculate ID

$$i_D = K_n (V_{GS} - V_{TN})^2$$

(a) $$V_G = \left(\frac{R_2}{R_1 + R_2}\right)(10) - 5 = \left(\frac{40}{40+60}\right)(10) - 5 = -1 \text{ V}$$

$$I_D = \frac{V_S - (-5)}{R_S} = K_n (V_{GS} - V_{TN})^2$$

$$V_S = V_G - V_{GS}$$

$$(5-1) - V_{GS} = (0.5)(1)\left(V_{GS}^2 - 2V_{GS} + 1\right)$$

$$0.5V_{GS}^2 - 3.5 = 0 \quad V_{GS}^2 = 7 \quad V_{GS} = 2.646 \text{ V}$$

$$I_D = (0.5)(2.646 - 1)^2 \Rightarrow I_D = 1.354 \text{ mA}$$

$$V_{DS} = 10 - (1.354)(3) = 5.937 \text{ V}$$

Since the gate current is zero, the gate is at ground potential. The voltage at the source terminal is then $V_S = -V_{GS} = -2.614\,\text{V}$. The value of the source resistor is found from

$$R_S = \frac{V_S - V^-}{I_{DQ}} = \frac{-2.614 - (-5)}{0.5}$$

$$R_S = 4.77\,\text{k}\Omega$$

The voltage at the drain terminal is determined to be

$$V_D = V_S + V_{DS} = -2.614 + 4 = 1.386\,\text{V}$$

The value of the drain resistor is

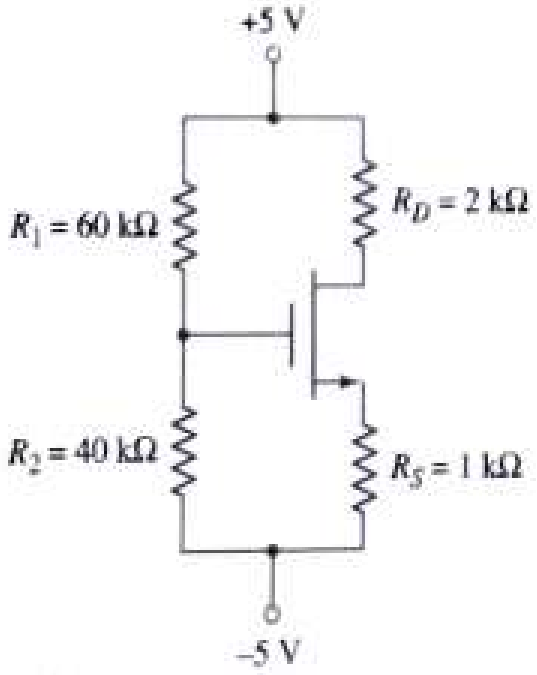
$$R_D = \frac{V^+ - V_D}{I_{DQ}} = \frac{5 - 1.386}{0.5}$$

We may note that

$$V_{DS} = 4\,V > V_{DS}(\text{sat}) = V_{GS} - V_{TN} = 2.61 - 1.2 = 1.41\,\text{V}$$

which means that the transistor is indeed biased in the saturation region.

Ex 3.5: (a) For the transistor in the circuit in Figure 3.29, the parameters are $V_{TN} = 1\,V$ a $K_n = 0.5\,mA/V^2$. Determine V_{GS}, I_D, and V_{DS}. (b) Determine the variation in the Q-point values for a ± percent variation in K_n and V_{TN}. (Ans. (a) $V_{GS} = 2.65\,V$, $I_D = 1.35\,mA$, $V_{DS} = 5.94\,V$; $1.297 \le I_{DQ} \le 1.411\,mA$, $5.768 \le V_{DSQ} \le 6.108\,V$)



$V_G = (R_2/(R_1+R_2))10 - 5 = -1\ V$

$I_D = [Vs - (-5)] / Rs = Kn\,(V_{GS} - V_{TN})^2$

$Vs = V_G - V_{GS}$

$(5-1) - V_{GS} = 0.5\,(V_{GS}^2 - 2\,V_{GS} + 1)$

$V_{GS} = 2.65\ V$

$I_D = Kn\,(V_{GS} - V_{TN})^2$

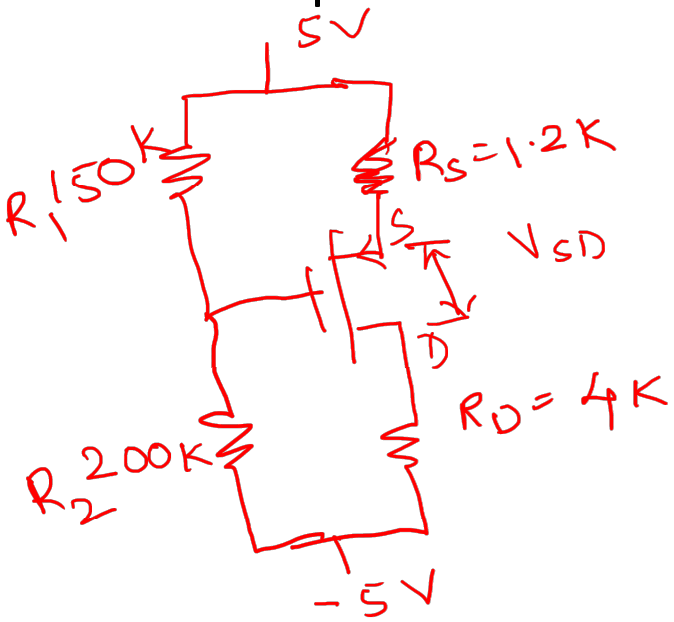
$V_{DS} = 10 - (1.35)(2+1) = 5.94\ V$

$(5-1) - V_{GS} = (1.05)\,0.5\,(V_{GS}^2 - 2\,V_{GS} + 1)$

$(5-1) - V_{GS} = (0.95)0.5\,(V_{GS}^2 - 2\,V_{GS} + 1)$

Ex 3.6

- R1 =150 K R2 = 200 k Rs = 1.2 K RD = 4 K Vss = 5 V VDD = -5 V
  PMOS potential divider biasing Kp = 0.25 mA/v^2 V_{TP} = -1 V



$$V_G = \left(\left(\frac{R_2}{R_1+R_2}\right)(5-(-5))\right) - 5$$

$$= \frac{200}{350}(10) - 5 = 0.7$$

$$V_S = 5 - I_D R_S =$$

$$I_D = K_P (V_{SG} + V_{TP})^2$$

$$V_{SG} = V_S - V_G = 5 - I_D R_S - 0.7$$

$$= 4.3 - I_D(1.2)$$

$$I_D = \frac{4.3 - V_{SG}}{1.2}$$

$$I_D = K_P (V_{SG} + V_{TP})^2$$

$V_{TP} = -1$

$$4.3 - V_{SG} = (1.2)(0.25)(V_{SG}^2 - 2V_{SG} + (-1)^2)$$

$$4.3 - V_{SG} = 0.3 V_{SG}^2 - 0.6 V_{SG} + 0.3$$

$$0.3 V_{SG}^2 + 0.4 V_{SG} - 4 = 0$$

-5.3

3.04

$$V_{SG} = 3.04\,V$$

$$I_D = K_D (V_{GG} + V_{TD})^2$$

$$= 0.25 (3.04 - 1)^2 = 1.04\,mA$$

$$V_{SD} = 10 - I_D (R_D + R_S) = 10 - (1.04)(1.2 + 4)$$

$$= 10 - (1.04 \times 5.2)$$

$$= 4.5$$

$V_{SD} > V_{SD(sat)}$

DC Equivalent Circuit

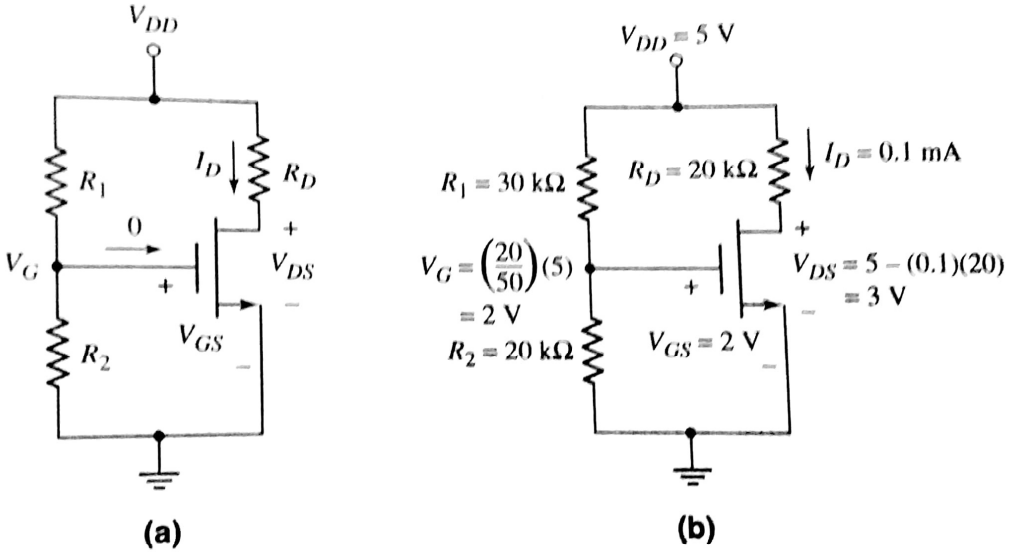


Figure 3.25 (a) The dc equivalent circuit of the NMOS common-source circuit and (b) the NMOS circuit for Example 3.3, showing current and voltage values

Reference

1. D. A. Neamen *Electronic Circuit Analysis and Design* ,McGraw Hill Education

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3. Thomas L. Floyd *Electronic Devices* , Prentice Hall (Pearson)